Logstash & Data *Ingestion*

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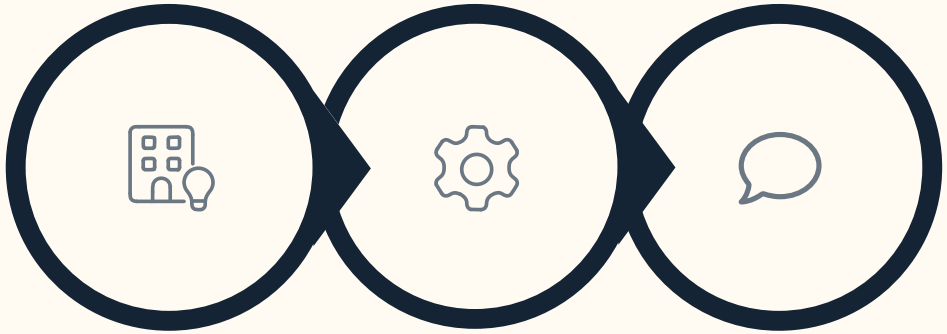
Learning Objectives

This module is designed to build practical skills in data ingestion and transformation using Logstash, enabling participants to build robust pipelines for Elasticsearch.



Understand Architecture

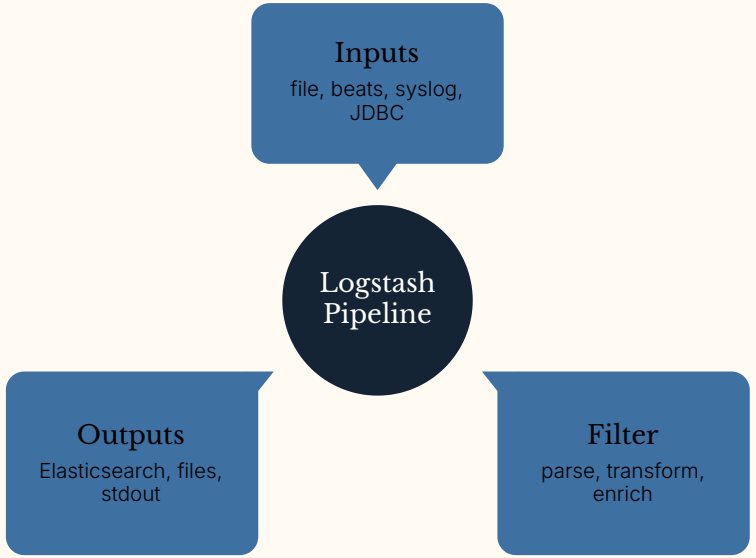
Configure Pipelines

Ingest to Elasticsearch

> Participants will gain hands-on experience building Logstash pipelines, applying Grok and mutate filters, and validating ingested data directly in Elasticsearch.

Logstash Overview

Logstash is a server-side data processing pipeline that ingests data from multiple sources, transforms it on the fly, and sends it to your desired destination — most commonly Elasticsearch.



Every stage of the pipeline can be **configured independently**, giving you full control over how data flows from source to destination.

Input Sources

- File & Beats
- Syslog
- JDBC

Output Targets

- Elasticsearch
- Files
- Stdout

Logstash plugins extend capabilities at every stage, making it adaptable to virtually any data source or destination.

Installation & Configuration

Install Logstash

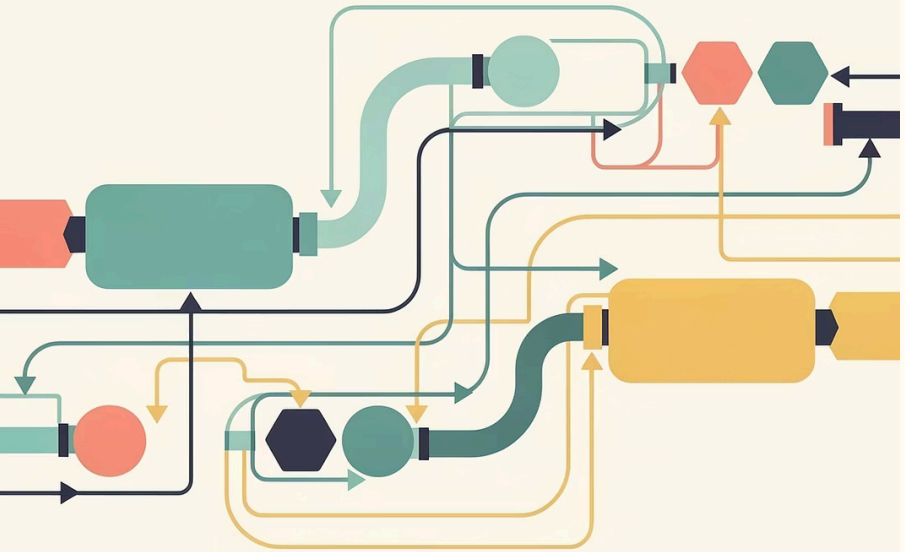
Download and install Logstash via APT, YUM, or direct download from the Elastic website. Ensure Java is available on the host system.

Write Pipeline Config

Create `.conf` pipeline files defining input, filter, and output sections. Store configs in the `pipeline/` directory or reference them directly.

Debug & Validate

Use `--config.test_and_exit` to validate syntax and `stdout { codec => rubydebug }` output to inspect processed events during development.



Data Transformation

Logstash provides a powerful set of **filter plugins** that allow you to parse, enrich, and reshape events before they reach Elasticsearch.

Grok Filter

Parse unstructured log data using pre-built or custom patterns. Ideal for Apache, Nginx, and syslog formats.

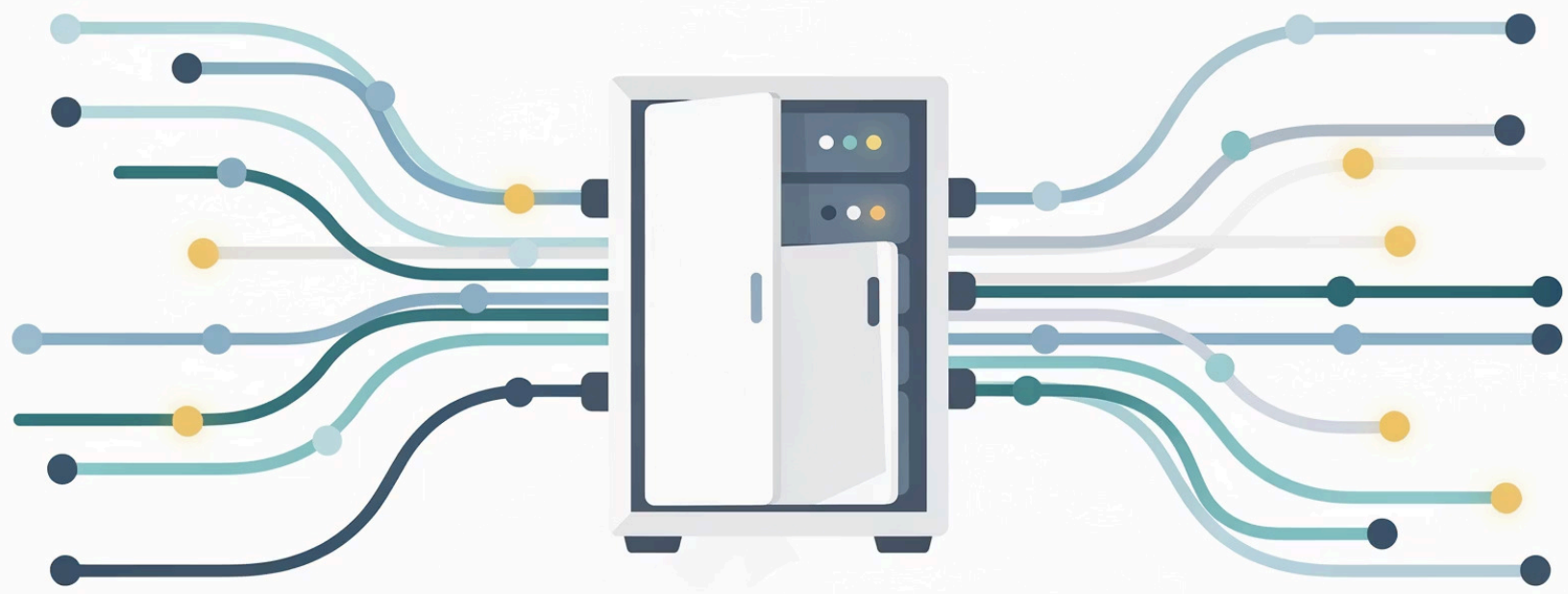
Date & Mutate

Normalize timestamps with the date filter and reshape fields — rename, remove, convert types — using the mutate filter.

Conditionals

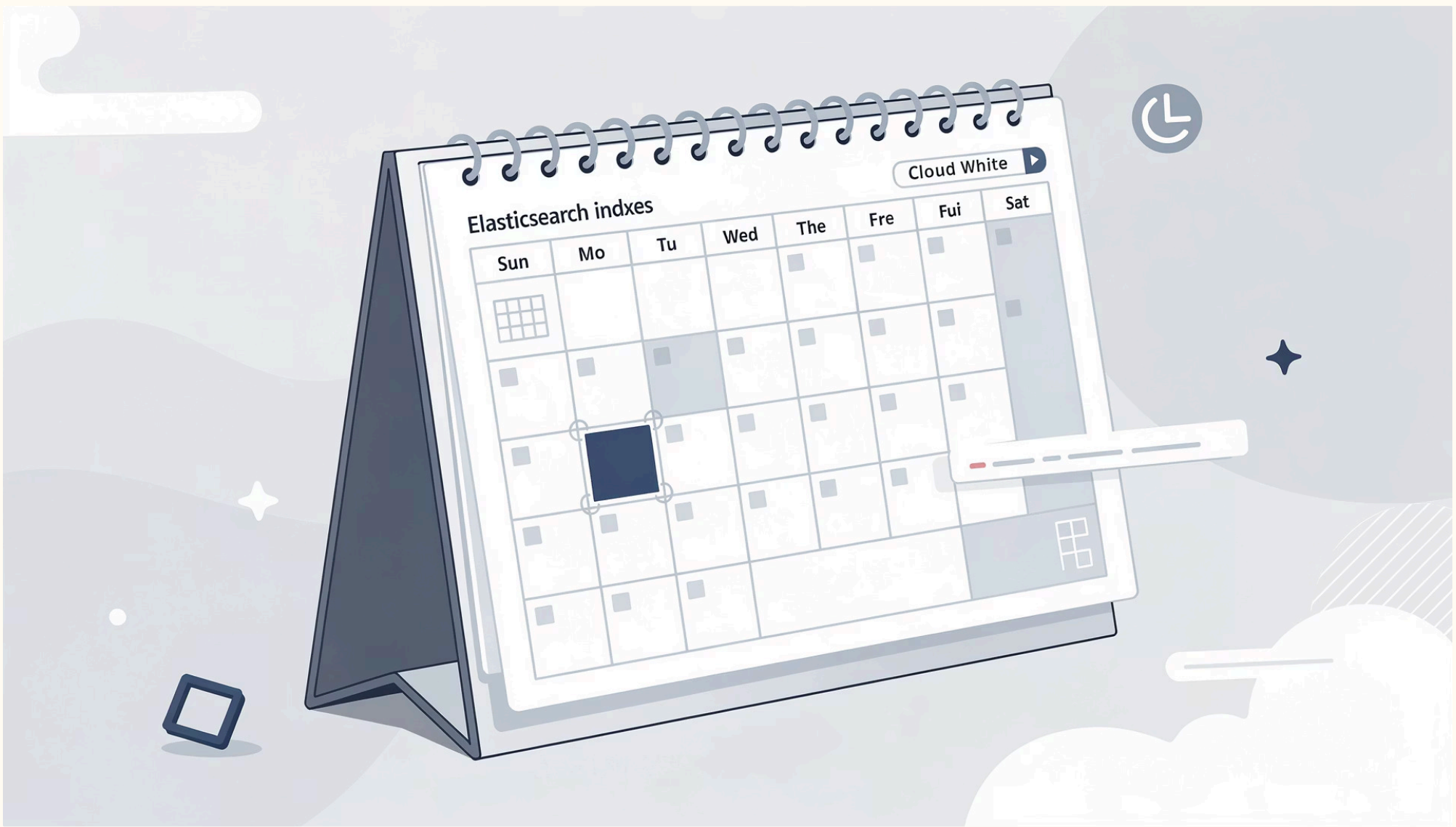
Use `if/else` conditionals in pipeline configs to apply filters selectively based on field values or event tags.

Combining Grok, date, and mutate filters gives you complete control over the structure and quality of data entering Elasticsearch.



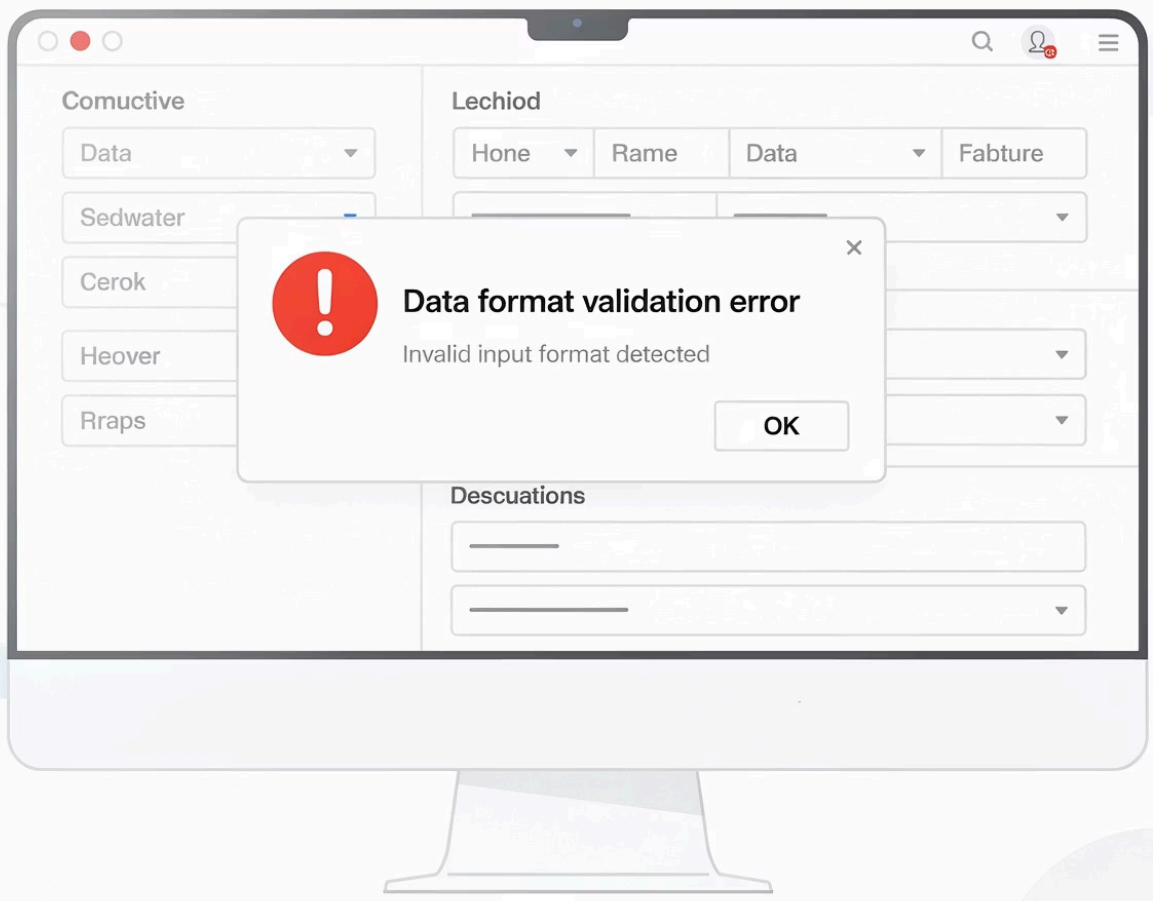
Send Data to Elasticsearch

Configure the Elasticsearch output plugin with your cluster hosts, index name, and authentication credentials to route processed events.



Dynamic Index Names

Use field interpolation such as `logstash-%{+YYYY.MM.dd}` to create time-based indices automatically, enabling efficient data management and ILM policies.



Handling Data Format Issues

Use `_grokparsefailure` tags and dead-letter queues to catch malformed events. Apply type coercion and field mapping to prevent index mapping conflicts.

Hands-On Labs

Lab 1: Apache Log Pipeline

Build a complete Logstash pipeline that reads Apache access logs from a file input and stores parsed events into an Elasticsearch index.

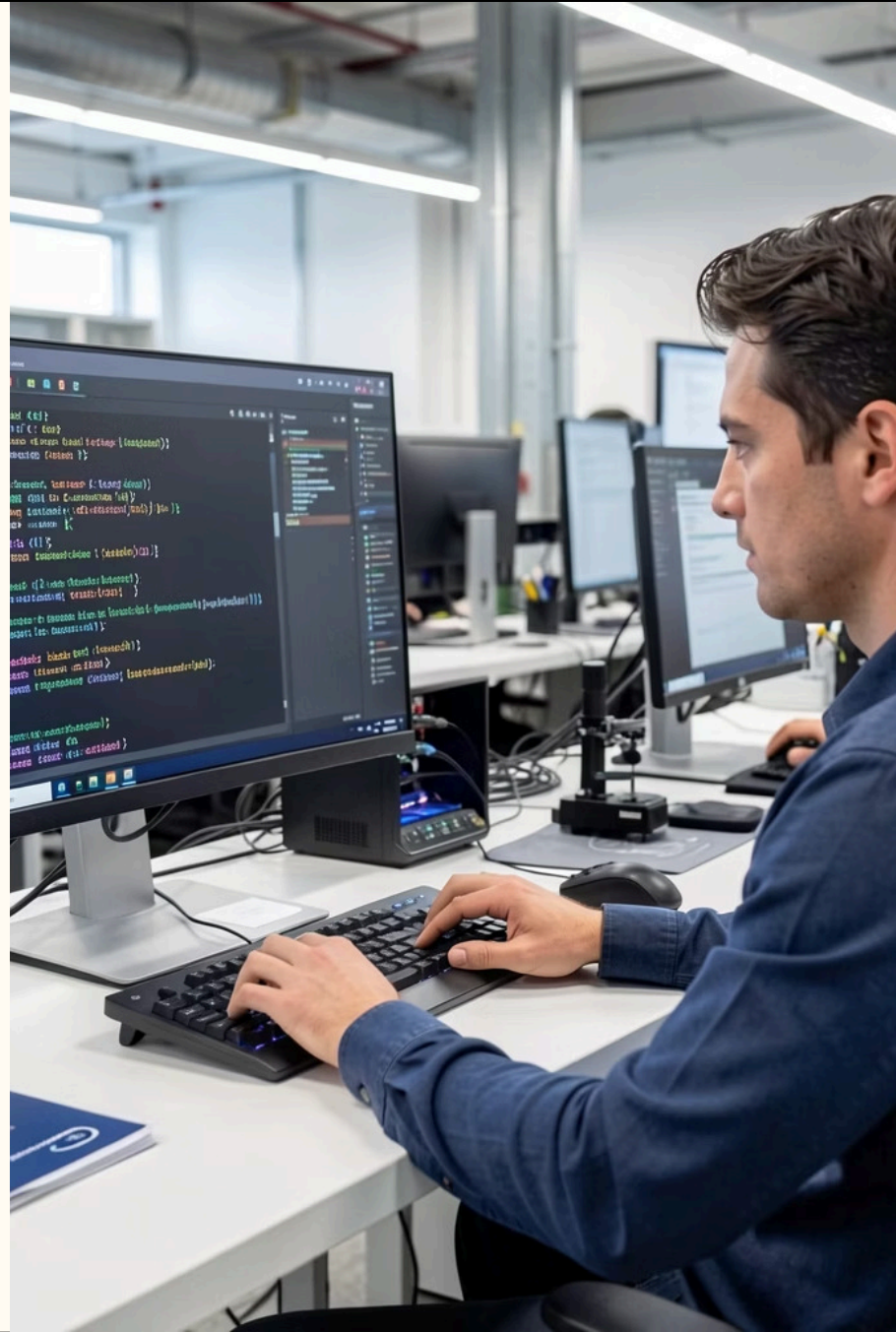
Lab 2: Grok & Mutate

Transform raw log data using Grok patterns for field extraction, date normalization, and mutate filters to rename and type-convert fields.

Lab 3: Validate in Elasticsearch

Use Kibana Dev Tools and Discover to query ingested data, verify index mappings, and confirm successful pipeline output.

Each lab builds on the previous one, reinforcing the full Logstash pipeline workflow from raw input to searchable Elasticsearch data.



Pipeline Configuration Reference

Input Block

Define your data source — file paths, Beats port, syslog listener, or JDBC connection string

Output Block

Route transformed events to Elasticsearch, a file, or stdout for debugging and verification

Filter Block

Apply Grok, date, mutate, and conditional logic to parse and enrich each event

Every Logstash pipeline follows the same three-block structure. Mastering each block is the foundation for building reliable, production-ready data ingestion workflows.

Tools & Resources

Official Documentation

The Elastic Logstash reference docs cover all input, filter, and output plugins with configuration examples and best practices for production deployments.

Instructor-Led Guidance

Experienced instructors from GYNOSIS INFORMATION TECHNOLOGY PRIVATE LIMITED guide participants through pipeline setup, debugging, and Elasticsearch integration challenges.

Lab Environment Access

Dedicated sandbox environments with pre-installed Logstash, Elasticsearch, and sample log datasets so participants can practice without any local setup overhead.

Core Logstash Principles

Building effective pipelines requires understanding the **fundamental principles** that govern how Logstash processes, transforms, and routes data at scale.

Event-Driven Processing

Every piece of data entering Logstash is treated as a discrete event that flows sequentially through input, filter, and output stages.

Plugin Extensibility

Hundreds of community and official plugins enable Logstash to connect to virtually any data source, apply complex transformations, and deliver to any destination.

Resilience & Reliability

Persistent queues and dead-letter queues ensure no data is lost during downstream failures, making Logstash suitable for production-grade pipelines.

What You Will Master